

AI FDE CAPSTONE

CGT Patient-to-Batch Consolidated Assessment

Context → Repository Assessment → Implementation → Evidence → Business Value →
Final Decision

Assessment position

A substantially completed synthetic academic FDE capstone, with production closure deliberately left open. The remaining gap is primarily proof, external assurance and field evidence - not another architecture rewrite.

Prepared for capstone review

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01 / DECISION

Executive Conclusion

Two different questions are being mixed together. Separating them resolves most of the disagreement around whether this is a capstone.

Question	Assessment
Has the challenge brief been substantially delivered as an academic FDE capstone?	YES - approximately 83% complete; 8.2/10.
Is this a field-deployed, independently assured, production FDE solution?	NO - production readiness is approximately 40-45%.

Why the difference matters

The repository goes well beyond a prototype: brownfield discovery, domain modelling, requirements, architecture, ADRs, three runnable workflow POCs, TEVV, security analysis, operational material, CI, traceability and a 90-day roadmap. But much of the final third of the FDE lifecycle is internal or simulated evidence rather than real-world evidence.

Recommended description

Use this wording

A substantially completed synthetic academic FDE capstone, with production closure deliberately left open.

End-to-end flow

1. CONTEXT	Brownfield CGT estate Fragmented systems + conflicting evidence + unsafe assumptions
2. ASSESSMENT	Forensic discovery → root causes → domain/data → requirements → architecture
3. IMPLEMENT	Deterministic orchestration + bounded AI + three vertical POCs
4. EVIDENCE	Tests + TEVV + traceability + security + runbooks + roadmap
5. DECIDE	Academic capstone substantially delivered; production closure remains open

02 / SOURCE CHALLENGE

1. Context

The challenge is not simply "build an application." It presents a deliberately messy multinational Cell & Gene Therapy patient-to-batch orchestration estate and explicitly asks the FDE to discover before designing.

What the challenge requires

#	Required outcome
1	Brownfield forensic assessment
2	Domain model + current-state journey reconstruction
3	Problem definition + measurable KPIs
4	Functional/NFR/assurance requirements
5	Target architecture + ADRs + migration
6	Three workflow POCs
7	TEVV including edge/adversarial/outage/authority
8	Security/privacy/Responsible AI/governance
9	Resilience/recovery/operational handover
10	Production-gap assessment + 90-day roadmap

Key challenge instruction

Do not assume every problem requires AI. Show where deterministic engineering is superior.

What this means for evaluation

The capstone should be judged on whether it demonstrates the FDE decision journey - not on how much AI was inserted.

- Understand ambiguity and field evidence
- Discover the real operational problem
- Model the domain and authority boundaries
- Decide where AI is justified - and where it is not
- Design and implement safe architecture
- Verify behavior with evidence
- Identify what blocks production
- Communicate the scale/change/stop decision

03 / EVIDENCE POSITION

2. Assessment of the Repository

Consolidated scorecard

Area	Assessment
Challenge deliverables represented	10 / 10
21 FDE stages represented	21 / 21
Canonical artifact register	185 artifacts
Missing canonical artifact files	0
Automated tests reported	92 passed
Requirements internally verified	29 / 31
Evaluation cases	57 executed
Three POCs	Implemented
CI	Implemented
Production deployment	No
Independent assurance	No
Real ROI/business outcome proof	No
Overall capstone completion	~83%
Repository assessment	8.2 / 10

Important boundary

The repository itself correctly states that this is a synthetic academic demonstration. Real deployment, owner approvals, independently assured outputs and observed business benefits are not implied.

04 / TEN OUTCOMES

Assessment by Client Deliverable

Requirement	Position	Why
Brownfield forensic assessment	Strong / delivered	Forensic baseline, inventory and evidence are present; real source-owner confirmation remains external.
Domain + current journey	Partial-strong	Domain model exists; six source-timestamp examples are reconstructed rather than full governed migration.
Problem + KPIs	Partial-strong	Problem/root cause is strong; some business denominators and real outcome baselines remain missing.
Requirements	Strong	31 canonical requirements; 29 internally verified, two need external human evidence.
Architecture + ADRs + migration	Strong design	Target C4, ADRs and migration plan exist; real owner approval/cutover proof absent.
Three workflow POCs	Delivered	Runnable identity/readiness, reservation/reconciliation and Quality/release verticals.
TEVV	Partial-strong	Broad catalog exists; many cases are structural probes rather than full property assertions.
Security/governance	Strong design; partial assurance	Threats/controls documented and locally tested; independent review absent.
Resilience/handover	Partial	Runbooks, restore and simulations exist; no real operations or representative DR.
Production roadmap	Delivered	Gap matrix and 90-day roadmap are explicit; open CAPAs are not falsely closed.

Where the maturity drops

The strongest work is approximately Stages 2-14. The gap begins when internal engineering evidence must become independent and operational evidence.

The critical distinction

"We designed it" → "We implemented it" → "We tested it internally" is not the same as "independent reviewers validated it" → "it ran against representative enterprise integrations" → "users adopted it" → "KPIs improved."

05 / MINIMUM VIABLE ARCHITECTURE

3. What Needed to Be Implemented

The correct implementation target was not a large multi-agent or RAG system. The smallest valuable solution is governed deterministic orchestration, with AI outside the consequential authority boundary.

Layer	Purpose
Patient / Batch Evidence	Preserve source-specific assertions and traceable history.
Governed Evidence Layer	Normalize evidence without inventing certainty.
Identity Authority	Human-authorized identity conflict resolution.
Readiness Engine	Deterministic prerequisite evaluation with explicit UNKNOWN.
Quality Authority	Prevent manufacturing status being treated as Quality release.
Command / Reservation Layer	Idempotency, query-before-retry, reconciliation and compensation.
Audit / Evidence	Tamper-evident, decision-linked evidence.
Optional AI	Recommendation-only; no consequential tool or authority.

Architecture judgement

A bounded deterministic design is stronger than forcing AI into patient identity, release or command authority. The challenge explicitly permits and encourages this decision.

What not to overengineer

- No giant multi-agent architecture
- No RAG just because the program is AI-focused
- No vector database without a retrieval problem
- No LLM authority over patient, Quality or workflow commands
- No microservice split simply to appear enterprise-grade

06 / ENGINEERING VALUE

The Three Workflow POCs

POC 1 - Identity → Readiness

Flow

Conflicting identity evidence → preserve conflict → create proposal → human-authorized decision → evidence-bound resolution → recalculate readiness.

Value: addresses wrong-patient/wrong-material linkage risk and prevents silent merging of conflicting identity evidence.

POC 2 - Reservation → Reconciliation

Flow

Reserve slot → timeout → OUTCOME_UNKNOWN → query downstream effect → reconcile → complete or compensate.

Value: demonstrates that a timeout is not necessarily a failure. Semantic idempotency, replay protection and reconciliation are more meaningful than a simple CRUD workflow.

POC 3 - Quality Evidence → Release

Rule

Manufacturing complete != product released. QC, deviations, thermal evidence and Quality authorization must be satisfied before release.

Value: protects a critical authority boundary and prevents operational status from being misinterpreted as Quality disposition.

AI boundary

AI may produce a cited recommendation schema. It has no consequential tool. AI-off is the default and the deterministic path remains complete.

07 / CODE + ARTIFACTS

4. What Has Actually Been Implemented

Functional implementation

Implementation	Evidence
Application composition	src/fde_capstone/application.py
API	src/fde_capstone/api.py
CLI	src/fde_capstone/cli.py
Browser Control Tower	src/fde_capstone/web/
Persistent evidence/state	src/fde_capstone/storage.py
Identity/readiness workflows	src/fde_capstone/services/
Reservation/reconciliation	Command service
Quality/release	Quality service
Optional assistant	Recommendation-only adapter
Container	Dockerfile
CI	.github/workflows/ci.yml
Automated tests	tests/

CI evidence chain

Pipeline

pytest → test summary → requirement verification → evaluation harness → delivery evidence generation → final capstone verification.

This is evidence-oriented engineering rather than a demo-only repository.

08 / WHERE TO LOOK

Reference Artifacts

Purpose	Repository reference
Challenge definition	source_baseline/participant/CHALLENGE_BRIEF.md
Final capstone summary	docs/FINAL_CAPSTONE_REPORT.md
Client acceptance matrix	docs/CLIENT_DELIVERABLE_ACCEPTANCE_MATRIX.md
21-stage register	docs/21_STAGE_ARTIFACT_REGISTER.md/.csv
Requirements	requirements/requirements.csv
Traceability	requirements/TRACEABILITY_MATRIX.csv
PRD	docs/stages/ stage_13/06_PRODUCT_REQUIREMENTS_DOCUMENT.md
Target C4	docs/stages/ stage_10/05_TARGET_C4_BASELINE_AND_AS_BUILT_DELTA.md
ADRs	docs/stages/stage_13/02_FINAL_ADRS.md
Migration	docs/stages/ stage_13/08_BROWNFIELD_MIGRATION_STRATEGY.md
Threat model	docs/stages/stage_12/01_THREAT_MODEL.md
TEVV	docs/stages/stage_15/01_TEVV_REPORT.md
Performance protocol	docs/stages/ stage_15/08_REGISTERED_JOURNEY_LOAD_TEST_PROTOCOL.md
Runbooks	docs/stages/stage_16/
Production roadmap	docs/stages/ stage_20/06_PRODUCTION_GAP_AND_90_DAY_ROADMAP.md
Progress status	docs/CAPSTONE_PROGRESS_TRACKER.md

Do not create more documentation

The repository already has enough documentation. The next improvement should be assessor navigation and stronger evidence, not another large set of Markdown files.

5. Business Value

1. Patient safety and decision integrity

Sources retain assertions; conflicts stay visible; evidence remains traceable; UNKNOWN remains UNKNOWN. This reduces false certainty entering a critical workflow.

2. Reduced duplicate operational effects

Idempotency and reconciliation prevent blind retry behavior when downstream outcomes are uncertain, potentially reducing duplicate reservations and manual recovery work.

3. Governance and accountability

Consequential decisions remain with defined human authorities - identity, Quality and controlled approvals - rather than an LLM.

4. Operational transparency

The Control Tower unifies patient state, batch state, readiness, exceptions, evidence, decisions and next actions into one operational view.

5. Reusable platform capability

The reusable value is the pattern: evidence model, assertions, authority boundaries, explicit uncertainty, idempotent commands, reconciliation and deterministic/AI separation.

Business-value boundary

Technical feasibility and risk-control value have been demonstrated. Measured business realization has not yet been established. Do not claim ROI, clinical benefit, reduced vein-to-vein time or quantified labor savings without real before/after data.

6. Documentation Flow for Review

A. Start here - five documents

1. README.md
2. source_baseline/participant/CHALLENGE_BRIEF.md
3. docs/FINAL_CAPSTONE_REPORT.md
4. docs/CLIENT_DELIVERABLE_ACCEPTANCE_MATRIX.md
5. docs/21_STAGE_ARTIFACT_REGISTER.md

These answer: what was asked → what was discovered → what was built → what is evidenced → what remains open.

B. Architecture / engineering

Stage	Use when challenged on
Stage 10	Target C4 / application and AI architecture
Stage 13	PRD, ADRs, requirements, migration and traceability
Stage 14	Implementation, POCs, source and tests

C. Assurance

Evidence	Purpose
Stage 7	Evaluation definition and thresholds
Stage 12	Threat model and security controls
Stage 15	TEVV report
requirements/	Verification matrix
tests/	Automated behavior evidence
.github/workflows/ci.yml	CI evidence

D. Production readiness

Stage 16 runbooks/recovery → Stage 17 deployment simulations → Stage 18 monitoring/resilience → Stage 20 gap matrix/CAPA/90-day roadmap.

Current release answer

No - this is not ready for real patient workflow or production deployment. The current lifecycle decision is RESTRICT_AND_CHANGE.

7. Has It Met the Capstone Requirement?

Why it is a defensible capstone

The challenge brief defines ten required deliverables, and the repository provides substantive evidence across all ten. It also demonstrates the core FDE loop:

FDE loop

Ambiguous brownfield problem → forensic discovery → problem qualification → domain understanding → requirements → architecture decisions → engineering → verification → operational gaps → business decision.

The strongest architectural judgement is the decision to keep AI outside consequential patient and Quality authority boundaries.

Why the team can still reasonably disagree

Concern	Assessment
1. Documentation-heavy appearance	21 stages and 185 artifact rows can make the project look like paperwork around a demo. The answer is stronger demo/evidence navigation - not more documents.
2. No real deployment	Stages 17-19 are mainly simulated. If your team defines a capstone as a customer/environment deployment, this version does not satisfy that stricter definition.
3. Spec-driven sequence not fully historically provable	The repository now contains the expected artifacts, but some were consolidated after implementation. Artifact alignment is high; proof that development occurred strictly spec-first is weaker.
4. Final 20% is external	Owner sign-off, controlled policies, enterprise IAM, representative integrations, human studies, full load testing, independent assurance and real business KPIs remain open.

12 / DECISION TABLE

Final Classification

Question	Conclusion
Is it just a prototype?	No
Is it only documentation?	No
Are real workflows implemented?	Yes - three POCs
Does it follow the challenge brief?	Substantially yes
Does it map to the 21-stage FDE model?	Yes - strongly at artifact level
Is strict spec-first execution historically proven?	Partially
Is internal verification credible?	Yes - with limitations
Is independent assurance complete?	No
Is it production ready?	No
Are actual business outcomes proven?	No
Is it a defensible academic FDE capstone?	Yes
Is it a completed real-world FDE engagement?	No

Consolidated assessment

83% academic-capstone completion / 8.2 out of 10 remains reasonable. The remaining problem is predominantly proof, external assurance and field evidence - not missing core architecture.

How to present it

Avoid leading with "we completed all 21 stages." That invites a production-evidence challenge.

Recommended defence

The capstone demonstrates the full FDE decision journey on a synthetic brownfield CGT problem: forensic discovery, domain reconstruction, requirements and architecture, three implemented vertical POCs, traceable TEVV, and an evidence-based decision not to progress to production until specific external controls are closed.

13 / SMALLEST EXECUTABLE CLOSE-OUT

What to Do Next

Do not continue building more architecture documentation. The highest-value work is stronger proof.

Priority	Action	Why
P0	Produce a 10-15 minute end-to-end demo proving all three POCs	Makes implementation undeniable.
P0	Create one challenge → requirement → architecture → code → test → result trace	Proves the FDE chain rather than document volume.
P0	Run the registered full journey/load test or explicitly disposition it	Closes the largest technical evidence gap.
P0	Convert critical TEVV structural probes into true property assertions	Moves from "ran" to "proved."
P0	Obtain one independent reviewer assessment	Reduces self-assessment criticism.
P1	Add Capstone Owner acceptance	Establishes accountable completion.
P1	Create a one-page before/after operational hypothesis	Makes value visible without fabricating ROI.

Target state

From debatable to defensible

The core implementation does not need another architectural rewrite. Complete the evidence loop and make the trace from challenge to working behavior obvious to an assessor.

Source Basis

This assessment consolidates the reviewed CGT capstone challenge package, the attached AI FDE End-to-End Capability Stack / 21-stage Operating Model, and the current FDE-Final-Capstone repository evidence reviewed on 16 September 2026. Repository claims are treated as internal evidence unless independently or operationally proven.